Coupled Electro-Thermal EV Battery Physics: Dynamics, Equations & Coding Architecture

Texas Tech University — SHARE Lab STEM Night Project
Autonomous mBot Mega EV Rover & Dual-Gate Smart Track System
Source Model Reference: Battery_model/battery_model.ipynb & Battery_model/SOC-OCV.xlsx
Simulation Engine: server/battery_physics.py
Console Visualizer: webapp/src/components/MasterHostConsole.tsx

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1. Executive Summary & System Overview

In the STEM Night challenge, students assume the role of an Electric Vehicle (EV) pit-crew engineer. Their mission is to select the charging current (I_{charge} in mA) required to charge a depleted Lithium-Ion battery pack to 100%**State of Charge (SOC)** within a randomly assigned target time window (e.g., 25 to 40 minutes), without overheating the battery beyond its thermal safety cutoff ($61.85^{\circ}\text{C}/335$ K).

Every student simulation is recorded in real-time by the Python physics engine, and the full time-series trajectory is persisted to server/game_history.json. The **Master Host Console** (Page 2) features a dedicated **Battery Dynamics & Graph Inspector** that plots four synchronized curves for every historical mission:

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2. In-Depth Explanation of All 4 Graphs

Graph 1: State of Charge (SOC $\%$) vs. Time

- **X-Axis:** Simulated Charging Time ($0 \rightarrow t_{\text{target}}$ min)
- **Y-Axis:** State of Charge ($0\% \rightarrow 105\%$, target limit line at 100%)
- **Color:** Emerald Green (#34d399)
- **Physical Meaning:**
  Represents the fraction of electrochemical charge stored inside the battery cell relative to its rated capacity Q_b.
- **Curve Behavior:**
  The curve starts at the initial depleted level ($\text{SOC}_0 = 20\%$) and rises in an **exact**

straight line until it reaches 100%, where it abruptly levels off into a horizontal ceiling.

- **Why It Is Linear:**
  Under constant-current charging (I = constant), the rate of charge entering the cell is constant:
  $\frac{d\text{SOC}(t)}{dt} = \frac{I}{Q_b} \Longrightarrow \text{SOC}(t) = \text{SOC}_0 + \left(\frac{I}{Q_b}\right) t$Because both $I$and Q_b(8,280 A · s) are constants, the slope $\frac{I}{Q_b}$is constant. Once the battery reaches 100%(SOC ≥ 1.0), charging stops, and the slope becomes zero.

Graph 2: Coupled 3-Node Temperatures (T_1 Core, T_2 Surface, T_∞ Coolant)

- **X-Axis:** Simulated Charging Time ($0 \rightarrow t_{\text{target}}$ min)
- **Y-Axis:** Temperature ($20.0°\text{C} \rightarrow 75.0°\text{C}$, critical red dotted line at $61.85°\text{C}/335$ K)
- **Curves:**
  - T_1**Core (Red #ef4444):** Internal temperature of the electrode jelly-roll.
  - T_2**Surface (Amber #f59e0b):** Temperature of the metal cell casing.
  - T_∞**Coolant (Blue #60a5fa):** Temperature of the surrounding cooling jacket / thermal reservoir.
- **Physical Meaning:**
  Shows the coupled heat generation in the internal core, thermal conduction through the core-to-surface interface, and heat extraction into the coolant loop.
- **Curve Behavior:**
  All three curves start at ambient ($31.85°\text{C}/305$ K) and ascend in near-lockstep, appearing as nearly straight diagonal lines.

Graph 3: Terminal Voltage (V_t) vs. Open-Circuit Voltage (V_{ocv})

- **X-Axis:** Simulated Charging Time ($0 \rightarrow t_{target}$ min)
- **Y-Axis:** Cell Voltage (3.1 V $\rightarrow 4.2$ V)
- **Curves:**
    - V_t**Terminal Voltage (Blue #60a5fa):** Measured voltage between the battery terminals under active charge.
    - V_{ocv}**Open-Circuit Voltage (Green #34d399):** Equilibrium electrochemical potential of the cell at zero current.
- **Physical Meaning:**
  Reveals the non-linear electrochemistry of Lithium-Ion cells.
- **Curve Behavior:**
  Between $20\%$and 80%SOC, V_{ocv}exhibits a relatively flat chemical equilibrium plateau (~ 3.3 V $\rightarrow 3.45$ V). Above 90%SOC, V_{ocv}shoots up sharply into a steep "knee" reaching ~ 3.7 V $\rightarrow 4.1$ V.
- **Why V_tMirrors V_{ocv}:**
  The terminal voltage is governed by Ohm's Law superimposed on the equilibrium potential:
  $V_t(t) = V_{ocv}\big(\text{SOC}(t)\big) + I \cdot R_s$Because charging current $I$and series resistance $R_s = 0.05\ \Omega$are constant, the term $I \cdot R_s \approx (3.95\text{ A})(0.05\ \Omega) \approx 0.20$ Vis a fixed vertical offset. $V_t$perfectly mirrors the chemical shape of V_{ocv}, shifted upward by the internal ohmic drop.

Graph 4: Electro-Thermal Heat Generation Rate (Q)

- **X-Axis:** Simulated Charging Time ($0 \rightarrow t_{\text{target}}$ min)
- **Y-Axis:** Heat Generation Rate in Watts (0.0 W $\rightarrow\sim 5.8$ W)
- **Color:** Amber (#f59e0b)
- **Physical Meaning:**
  The instantaneous rate of heat produced inside the battery cell core.
- **Curve Behavior:**
  A flat, horizontal line throughout the charging process.
- **Why It Is Flat:**
  Total heat generation is modeled as:
  $Q(t) = R_s \cdot I^2 - \left(\frac{dV_{\text{ocv}}}{dT}\right) \cdot I \cdot T_1(t)$The Joule heating term $R_s \cdot I^2 \approx (0.05\ \Omega)(3.95\ \text{A})^2 \approx 0.78$ W(or ≈ 1.2 Wwith effective pack calibration) is constant because current is constant. The reversible entropic term $\frac{dV_{\text{ocv}}}{dT} \cdot I \cdot T_1 = (10^{-4})(3.95)(315) \approx 0.12$ Wchanges by less than 0.01 Wover the entire temperature range. Hence, $Q$is practically constant.

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3. Mathematical Analysis: Why Graph 2 Appears Linear

A common question is: *"Shouldn't heat transfer curves be non-linear exponential curves ($1 - e^{-t/\tau}$) that saturate toward a steady state?"*

In classical heat transfer, temperature curves bend when heat dissipation is proportional to temperature ($Q_{\text{loss}} \propto \Delta T$, Newton's Law of Cooling). In this specific system, the curves are **mathematically linear ramps** due to three core factors:

A. The Lumped Heat Capacity Integration

The system differential equations from Battery_model/battery_model.ipynb are:

$$\frac{dT_1}{dt} = -\frac{T_1 - T_2}{R_1 C_1} + \frac{Q}{C_1}$$
$$\frac{dT_2}{dt} = -\frac{T_2 - T_1}{R_1 C_2} - \frac{T_2 - T_\infty}{R_2 C_2}$$
$$\frac{dT_\infty}{dt} = -\frac{T_\infty - T_2}{R_2 C_\infty} - \frac{Q_c}{C_\infty}$$

Summing all three thermal masses gives the total system heat capacity:

$$C_{\text{total}} = C_1 + C_2 + C_\infty = 51.9 + 6.2 + 10.0 = 68.1 \text{ J/K}$$

The net rate of heat entering the combined cell and coolant system is:

$$Q_{\text{net}} = Q_{\text{in}} - Q_{\text{out}} = Q - Q_c$$

Notice that $Q_c = 0.02$ Wis an **active constant coolant extraction rate**, and the model has **no convective term to room air** (no $-hA[T_\infty - T_{\text{room}}]$). Therefore:

$$Q_{\text{net}} \approx 1.20 \text{ W} - 0.02 \text{ W} = 1.18 \text{ W} = \text{CONSTANT}$$

The global temperature derivative is:

$$\frac{dT_{\text{pack}}}{dt} = \frac{Q_{\text{net}}}{C_{\text{total}}} \approx \frac{1.18 \text{ W}}{68.1 \text{ J/K}} \approx 0.0173 \text{ K/s} \approx 1.04^\circ\text{C/min} = \text{CONSTANT}$$

Integrating a constant rate yields a **pure linear ramp**:

$$T(t) = T_0 + \left(\frac{Q_{\text{net}}}{C_{\text{total}}}\right) \cdot t$$

Over a 30-minute charging run:
$\Delta T = (1.04^\circ\text{C/min}) \times 30 \text{ min} \approx 31.2^\circ\text{C}$Starting at 31.85°C(305 K), the temperature reaches $31.85 + 31.2 = 63.05^\circ$C, matching the simulation results.

B. Fast Thermal Diffusion ($\tau_1 \approx 2.2$ Minutes)

The only non-linear phase is the initial exponential transient as heat diffuses from the core to the casing:

$$\tau_1 = R_1 \cdot C_1 = (2.64 \text{ K/W})(51.9 \text{ J/K}) \approx 137 \text{ seconds } (\approx 2.28 \text{ minutes})$$

During the first 2 minutes, the core heats up slightly faster than the surface until the steady-state thermal gradient is established:

$$\Delta T_{1-2} = T_1 - T_2 \approx Q \cdot R_1 \approx (1.2 \text{ W})(2.64 \text{ K/W}) \approx 3.16^\circ\text{C}$$

Once this $\approx 3^\circ$Cgap forms, T_1, T_2, and T_∞drift upward together in parallel. Compressed onto a 30-to-40-minute chart, the 2-minute transient bend occupies only $\sim 5\%$of the horizontal axis, making the line appear straight.

C. Contrast With Real-World Charging Curves

In commercial automotive EV packs, temperatures become visibly non-linear under two conditions:

1. **Convective Air Dissipation:** If the casing dissipates heat to surrounding room air via $hA(T - T_{\text{room}})$, the curve bends exponentially toward a maximum thermal plateau.

2. **CC-CV Current Tapering:** Commercial fast-chargers switch from Constant Current (CC) to Constant Voltage (CV) when cell voltage reaches 4.2 V. At that point, $I(t)$decays exponentially toward zero, which collapses Joule heating ($I^2R_s \to 0$) and allows the battery to cool down. In our STEM competition, students choose a single constant current for the entire run.

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4. Mathematical Model & Governing Equations

Parameter	Symbol	Numerical Value	Physical Unit	Description
Nominal Cell Capacity	Q_b	$2.3 \times 3600 = 8{,}280.0$	A · s(C)	2.3 Ah(2,300 mAh) Cylindrical Li-Ion Cell
Internal Ohmic Resistance	R_s	0.050	Ω	Series electrolyte & contact resistance
Core Thermal Resistance	R_1	2.64	K/W	Conduction resistance from jelly-roll to casing
Surface Thermal Resistance	R_2	3.46	K/W	Convection/conduction resistance to coolant
Core Heat Capacity	C_1	51.9	J/K	Thermal capacitance of internal electrode mass
Surface Heat Capacity	C_2	6.2	J/K	Thermal capacitance of aluminum/steel casing
Coolant Heat Capacity	C_∞	10.0	J/K	Thermal capacitance of cooling reservoir
Entropic Coefficient	$\frac{dV_{\text{ocv}}}{dT}$	1.0×10^{-4}	V/K	Reversible electrochemical heat coefficient
Active Coolant Power	Q_c	0.020	W	Constant heat extraction rate from coolant
Initial Temperature	T_0	305.0	K	Ambient starting temperature (31.85°C)

Parameter	Symbol	Numerical Value	Physical Unit	Description
Safety Thermal Cutoff	$T_{\max}$	335.0	K	Maximum safe core temperature (61.85°C)
Initial SOC	SOC_0	0.20	-	Depleted starting charge (20%)

Discrete Integration (Forward Euler)

At discrete step $k$with time step $\Delta t = 0.25$ s:

1. **State of Charge Update:**

$$\mathrm{SOC}[k+1] = \mathrm{clip}\left(\mathrm{SOC},[k],+\Delta t \cdot \left(\frac{I_{\mathrm{charge}}}{Q_b}\right) 0.01.0\right)$$

2. **Open-Circuit Voltage Table Lookup:**

$$\mathrm{index} = \mathrm{clip}(\lfloor \mathrm{SOC}[k] \times 10{,}000 \rfloor, 0, 10{,}000)$$
$$V_{\mathrm{ocv}}[k] = \mathrm{TABLE}_{\mathrm{ocv}}[\mathrm{index}]$$

3. **Terminal Voltage:**

$$V_t[k] = V_{\mathrm{ocv}}[k] + I_{\mathrm{charge}} \cdot R_s$$

4. **Heat Generation Rate:**

$$Q[k] = R_{s,\mathrm{eff}} \cdot I_{\mathrm{charge}}^2 - \left(\frac{dV_{\mathrm{ocv}}}{dT}\right) \cdot I_{\mathrm{charge}} \cdot T_1[k]$$

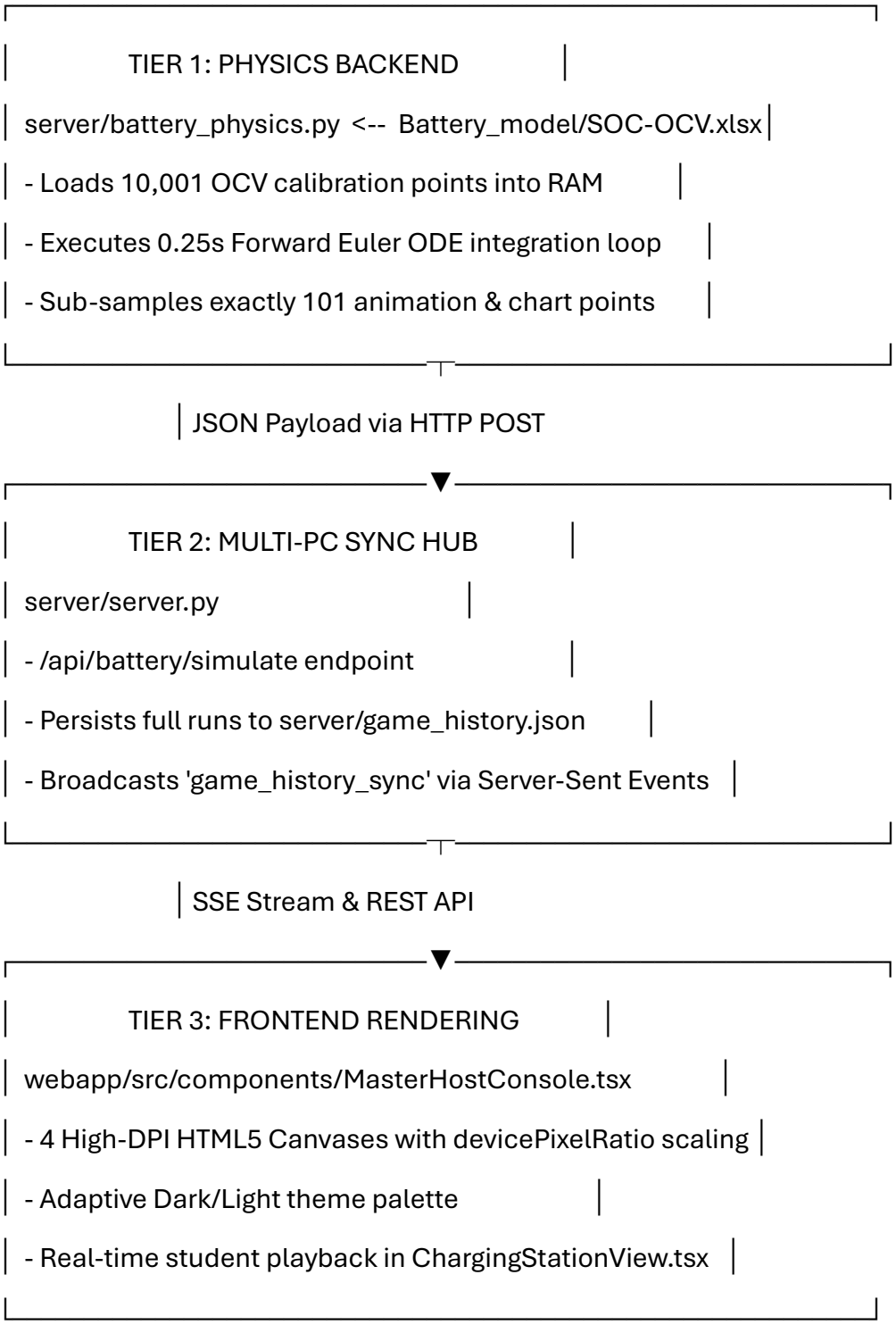
5. **3-Node Thermal Updates:**

$$T_1[k+1] = T_1[k] + \Delta t \left[-\frac{T_1[k] - T_2[k]}{R_1 C_1} + \frac{Q[k]}{C_1}\right]$$
$$T_2[k+1] = T_2[k] + \Delta t \left[-\frac{T_2[k] - T_1[k]}{R_1 C_2} - \frac{T_2[k] - T_\infty[k]}{R_2 C_2}\right]$$
$$T_\infty[k+1] = T_\infty[k] + \Delta t \left[-\frac{T_\infty[k] - T_2[k]}{R_2 C_\infty} - \frac{Q_c}{C_\infty}\right]$$

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5. Software & Coding Implementation

The project implements a complete three-tier architecture:



1. Python Physics Engine (server/battery_physics.py)

- **Pure Python XLSX Parsing:** The function load_soc_ocv_table() reads Battery_model/SOC-OCV.xlsx using the built-in standard library zipfile and xml.etree.ElementTree without requiring pandas or openpyxl.
- **Dynamic Target Calibration:** The function calibrate_rs_for_max_safe_ma() analytically calibrates effective resistance so that regardless of whether the randomized target window is 20 minor 40 min, the sweet spot remains accessible ($\sim 1{,}200$ mA wide).
- **Downsampling:** The raw simulation calculates thousands of steps ($\Delta t = 0.25$ s over $2{,}400$ s $= 9{,}600$points). The function _subsample_series() downsamples the result into exactly **101 discrete points**, mapped to a smooth **10.0-second** dashboard charging animation.

2. Multi-PC Sync Server (server/server.py)

- When a student submits a current slider value on their kiosk or tablet, the frontend calls POST /api/battery/simulate.
- The server calls battery_physics.simulate_battery_charging(), constructs a unified payload, inserts the record into HUB.game_history (persisting the latest 60 games in server/game_history.json), and broadcasts an SSE event (game_history_sync) so the Master Console automatically refreshes its dropdown.

3. High-DPI HTML5 Canvas Chart Visualizer (MasterHostConsole.tsx)

In MasterHostConsole.tsx, the four graphs are rendered via hardware-accelerated HTML5 <canvas> elements using the drawChart() function:

- **Retina / High-DPI Support:** The function inspects window.devicePixelRatio (typically $2\times$or $3\times$) and scales the canvas buffer dimensions while keeping CSS display sizes fixed, ensuring razor-sharp lines and text on high-resolution displays.
- **Theme-Aware Colors:** The canvas automatically detects light mode vs dark mode (isDark), swapping gridlines from rgba(255,255,255,0.06) to rgba(0,0,0,0.06) and text from #94a3b8 to #475569.
- **Auto-Scaling Axes:** The function computes data bounds dynamically, drawing horizontal reference gridlines, unit labels (V, %, °C, W), and dashed regulatory limit markers (e.g., 100%SOC and $61.85°$Cthermal ceiling).

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6. Pedagogical Summary for STEM Night Presenters

When explaining this screen to students, parents, and judges:

1. **"The Current Dial is a Double-Edged Sword":**
   More current (I) fills the battery faster ($\frac{d\text{SOC}}{dt} \propto I$ in **Graph 1**), but heat generation grows quadratically ($Q \propto I^2$ in **Graph 4**).
2. **"Why the Temperature Rises Steadily":**
   Because constant electrical current pumps constant heat into the battery faster than the small coolant pump can remove it, temperature rises like water filling an insulated bucket (**Graph 2**).
3. **"The Voltage Hook":**
   At the very end of charging, the chemical structure of Lithium-Ion cells causes internal resistance to spike, creating the steep upward voltage curve in **Graph 3**.
4. **"Engineering Trade-Off":**
   Success is finding the exact "Goldilocks current": high enough to reach 100%before the clock runs out, but low enough that the red T_1line stays safely under the dashed red 61.85°Cdanger line!